

A Course Based Project Report on
DISTRIBUTED CUSTOMER ANALYTICS

Submitted to the
Department of CSE-(CyS, DS) and AI&DS

in partial fulfilment of the requirements for the completion of course
BIG DATA COMPUTING LABORATORY (22PCDS303)

BACHELOR OF TECHNOLOGY

IN

CSE-Data Science

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CERTIFICATE

This is to certify that the project report entitled “**Distributed Customer Analytics**” is a bonafide work done under our supervision and is being submitted by **Mr. J. Jagadeeswar (23071A67F4), Mr. K. Parvez Alam (23071A67F5), Mr. K. Varshith Reddy (23071A67F7), Mr. M. Nehal Reddy (23071A67G1)** in partial fulfillment for the award of the degree of **Bachelor of Technology in CSE-Data Science**, of the VNRVJiet, Hyderabad during the academic year 2025-2026.

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DECLARATION

We declare that the course based project work entitled “**Distributed Customer Analytics**” submitted in the Department of **CSE - (CyS, DS) and AI&DS**, Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology, Hyderabad, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in CSE-Data Science** is a bonafide record of our own work carried out under the supervision of **Mrs. P. Swathi, Assistant Professor, Department of CSE-(CyS, DS) and AI&DS, VNRVJIET**. Also, we declare that the matter embodied in this thesis has not been submitted by us in full or in any part thereof for the award of any degree/diploma of any other institution or university previously.

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ABSTRACT

The **Distributed Customer Analytics project** leverages the Hadoop ecosystem to analyze large-scale user behavior data, enabling the identification of patterns and trends that are critical for improving customer engagement and overall user experience. The system is designed as a complete data pipeline that ingests structured data from traditional relational databases, transfers it into a distributed environment, and processes it efficiently using parallel computation techniques.

At the storage layer, the **Hadoop Distributed File System (HDFS)** is used to store data across multiple nodes, ensuring fault tolerance and scalability through block replication. For data ingestion, **Sqoop** acts as a bridge between relational databases (such as MySQL) and HDFS, enabling efficient and parallel transfer of structured data into the Hadoop ecosystem. Once the data is available in HDFS, it becomes accessible for distributed processing.

The processing layer utilizes **MapReduce**, which breaks down large datasets into smaller chunks and processes them in parallel across multiple nodes, significantly reducing computation time. On top of this, **Hive** provides a high-level SQL-like interface that allows users to perform complex analytical queries without writing low-level code, translating these queries into execution jobs internally. In addition, **Pig** is used for data transformation and ETL operations, offering a procedural scripting approach to clean, filter, and aggregate data before analysis.

By integrating these components, the project demonstrates a complete end-to-end big data workflow, from data ingestion and storage to processing and querying. The final output of the system consists of meaningful insights, such as customer segmentation, purchasing behavior patterns, and aggregated metrics, which can be used to support data-driven decision-making in real-world business scenarios. Overall, this project highlights how distributed systems can efficiently handle and analyze massive datasets that are otherwise impractical to process using traditional single-node systems.

Keywords: Hadoop, HDFS, MapReduce, Hive, Pig, Sqoop.

CHAPTER-1

INTRODUCTION

In the digital age, understanding user behavior is paramount for businesses aiming to stay competitive in an increasingly data-driven environment. Every interaction—whether it is a click, purchase, search query, or time spent on a platform—generates valuable data that can reveal patterns, preferences, and trends. However, the sheer volume, velocity, and variety of this data make it difficult to process using traditional systems. This is where scalable and distributed technologies become essential. Organizations need robust tools and frameworks that can efficiently handle massive datasets while enabling real-time or near real-time insights. Without such systems, valuable information remains untapped, leading to missed opportunities in personalization, optimization, and strategic decision-making.

The Hadoop ecosystem provides a powerful and reliable solution for addressing these challenges. With its distributed storage and parallel processing capabilities, Hadoop allows data to be stored across multiple nodes and processed simultaneously, significantly improving performance and scalability. In this project, we will design and implement a comprehensive data analysis pipeline specifically focused on user interaction data. The goal is to not only store and manage large volumes of data but also to transform raw data into actionable insights that can drive business decisions.

To achieve this, we will first set up the Hadoop Distributed File System (HDFS), which will serve as the backbone for storing large-scale datasets in a fault-tolerant and distributed manner. Once the data is stored, we will use MapReduce to process and analyze it by breaking down complex computations into smaller, parallel tasks. For higher-level querying and easier data interaction, we will integrate Hive, which provides an SQL-like interface for querying large datasets stored in HDFS. Additionally, Pig will be used for scripting and performing data transformations, making it easier to clean, filter, and prepare data for analysis.

CHAPTER-2

METHODOLOGY

2.1 Overview and Architecture

Data Collection and Storage:

- **HDFS:** User interaction data is ingested and stored in the Hadoop Distributed File System (HDFS). Below is an example of how data might be stored in HDFS using Hadoop.

Commands:

```
hadoop fs -put /local/path/to/user_interaction_data /user/hadoop/data/
```

Data Integration:

- **Sqoop:** Data from existing SQL databases is imported into HDFS using Sqoop. Here's a snippet demonstrating Sqoop usage to import data from MySQL:

```
sqoop import connect jdbc:mysql://mysql_server/blogspot_db -username user  
password pass -table user_interactions -target-dir/user/hadoop/data/
```

Data Processing:

- **MapReduce:** MapReduce jobs are implemented for processing large datasets in a distributed manner.

Here's a simplified example of a MapReduce program to count user interactions:

```
public class UserInteractionCount {  
    public static void main(String[] args) throws Exception {  
        Job job = Job.getInstance(new Configuration(), "User Interaction Count");  
        job.setJarByClass(UserInteractionCount.class);  
        FileInputFormat.addInputPath(job, new Path("/user/hadoop/data/user_interactions"));  
        FileOutputFormat.setOutputPath(job, new  
Path("/user/hadoop/output/interaction_count"));  
        job.setMapperClass(InteractionMapper.class);  
        job.setReducerClass(InteractionReducer.class);  
        job.setOutputKeyClass(Text.class);  
        job.setOutputValueClass(IntWritable.class);  
        System.exit(job.waitForCompletion(true) ? 0 : 1);  
    }  
}
```



```
}  
}
```

2.2 Implementation Steps

Data Querying:

- **Hive:** Hive queries are used for data querying and analysis. Here's an example of a Hive query to find the average user interaction duration:

```
SELECT user_id, AVG(duration) AS avg_duration  
FROM user_interactions  
GROUP BY user_id;
```

Data Transformation:

- **Pig:** Pig scripts are utilized for data transformation tasks. Below is an example of a Pig script to filter and transform user interaction data:

```
user_interactions = LOAD '/user/hadoop/data/user_interactions' USING  
PigStorage(',') AS (user_id:int, action:chararray, timestamp:long);  
filtered_interactions = FILTER user_interactions BY action == 'click';  
grouped_by_user = GROUP filtered_interactions BY user_id;  
user_interaction_count = FOREACH grouped_by_user GENERATE group AS  
user_id, COUNT(filtered_interactions) AS click_count;  
STORE user_interaction_count INTO '/user/hadoop/output/user_interaction_count'  
USING PigStorage(',');
```

Data Analysis and Visualization:

Insights extracted through Hive queries and Pig scripts can be visualized using tools like Tableau or custom dashboards. Visualization aids in interpreting user behavior patterns effectively.

CHAPTER-3

TEST CASES/ OUTPUT

Test Case-1:

```
1, Alice, 30, Female, 150.00,2024-01-01
2, Bob, 25, Male, 200.50, 2024-02-15
3, Charlie, 35, Male, 300.75,2024-03-20
```

Test Case-2:

```
customer_data_hive
```

Test Case-3:

```
1  Alice      30  Female  150.00  2024-01-01
2  Bob        25  Male    200.50  2024-02-15
3  Charlie    35  Male    300.75  2024-03-20
```

Test Case-4:

```
Female  150.00
Male    250.625
```

CHAPTER-4

RESULTS

Import customer data from MySQL. to HDFS.

```
sqoop import
--connect jdbc:mysql://localhost:3306/customer_db
--username your_username
--password your_password
--table customer_data
--target-dir/user/hadoop/customer_data
--m 1
```

Result: Data was successfully imported to HDFS

Analyze customer data to derive insights such as average purchase amount by gender.

```
SELECT gender, AVG(purchase_amount) AS avg_purchase
FROM customer_data_hive
GROUP BY gender;
```

Result: Successfully derived insights from the customer data.

Female	150.00
Male	250.625

CHAPTER-5

DISCUSSION

1. Hadoop Distributed File System (HDFS):

- HDFS serves as the foundational storage layer in the project. It accommodates large volumes of user interaction data in a distributed manner, ensuring reliability and scalability. This capability is crucial as it allows the project to handle vast amounts of data generated from user interactions across various channels.

2. MapReduce:

- MapReduce is employed for data processing tasks within the project. It facilitates parallel processing of large datasets across a distributed network of nodes. By breaking down complex data processing tasks into smaller manageable tasks (maps and reduces), MapReduce enables efficient computation and aggregation of user behavior patterns and trends.

3. Hive:

- Hive is utilized for querying and analyzing the processed data stored in HDFS. Its SQL-like interface simplifies data exploration and extraction of insights from structured data sets. This capability is instrumental in enabling analysts and data scientists to derive actionable insights into customer behavior through ad-hoc queries and data summarization.

4. Pig:

- Pig complements MapReduce by offering a high-level data flow language. It allows for the scripting of data transformations and manipulations that are executed on Hadoop. This flexibility is valuable in handling diverse data processing requirements, such as cleansing, transforming, and preparing data for analysis.

5. Sqoop:

- Sqoop facilitates seamless data integration between traditional SQL databases and the Hadoop ecosystem. It supports the efficient import and export of data, ensuring that all relevant data sources are accessible for analysis within the Hadoop environment. This capability enables the project to leverage existing data assets from relational databases, enhancing the comprehensiveness of insights derived from user behavior data.

CONCLUSION

The Customer Insights Hub project effectively showcases the strength and practicality of the Hadoop ecosystem in handling and analyzing large-scale user behavior data. In modern systems, user-generated data is not only massive in volume but also highly dynamic, requiring a distributed architecture that can scale horizontally without compromising performance. This project addresses that challenge by building a pipeline that is both robust and extensible, ensuring that data from multiple sources can be ingested, stored, processed, and analyzed efficiently. By organizing the workflow into distinct layers—storage, processing, querying, and integration—the system reflects how real-world big data platforms are designed in production environments.

At the storage layer, Hadoop Distributed File System (HDFS) ensures fault-tolerant and distributed storage, allowing data to be split into blocks and replicated across nodes. This design guarantees high availability and reliability even in the case of node failures. On top of this storage layer, MapReduce provides the core processing mechanism, enabling parallel computation across clusters. Instead of processing data sequentially, tasks are divided into map and reduce phases, significantly improving throughput for large datasets. This makes it feasible to process logs, clickstreams, and behavioral data at scale without bottlenecks.

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